Experiment 4.3: Design and Analysis of an FIR Low-Pass Filter Using the Hanning Window Technique in MATLAB

Aim: To design and analyze a low-pass FIR filter using the Hanning window technique in MATLAB.

MATLAB Code

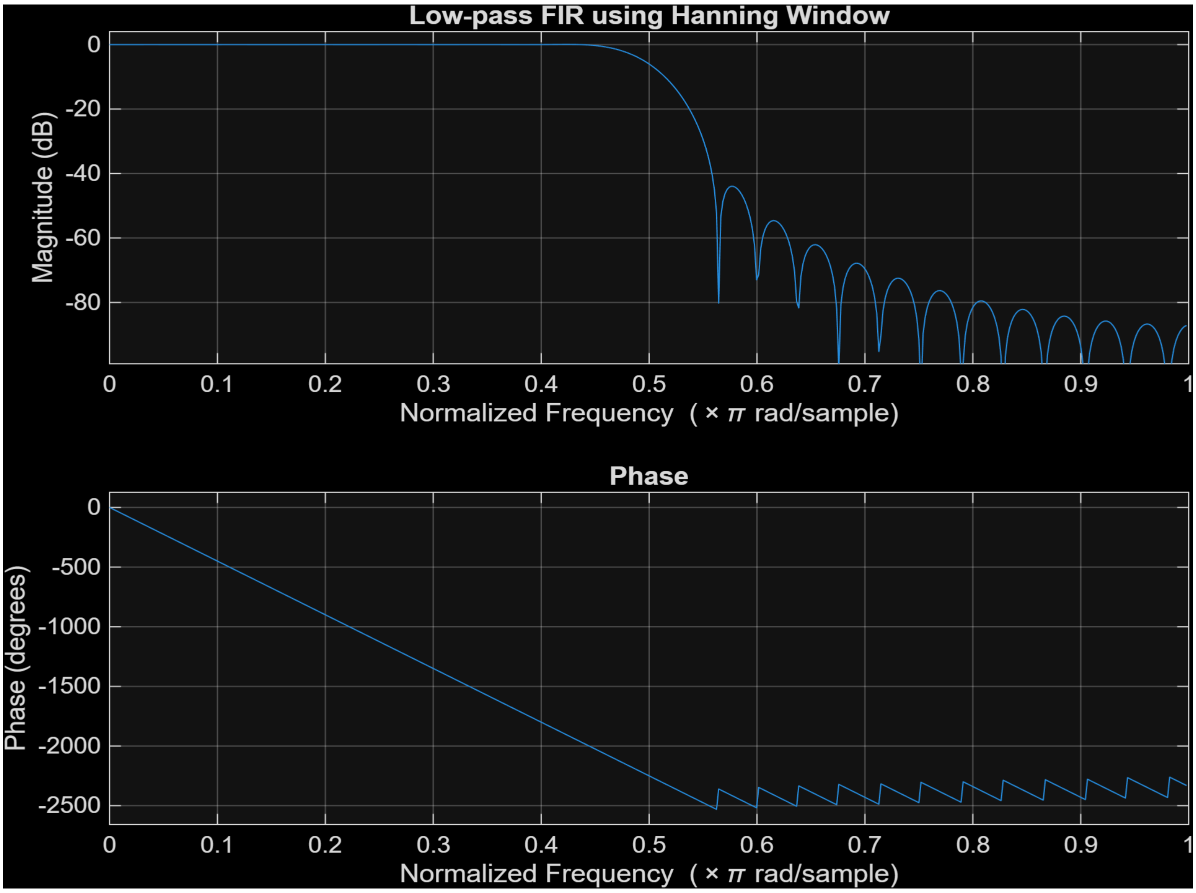
```
clc;
clear;
close all;

%% Save Command Window Output
diary('Exp4_3_Output.txt');

%% Input Parameters
fp = 1000;      % Passband cutoff frequency (Hz)
fs = 4000;      % Sampling frequency (Hz)
N = 50;         % Filter order

wc = fp/(fs/2); % Normalized cutoff frequency

%% FIR Low Pass Filter using Hanning Window
b_hann = fir1(N, wc, 'low', hanning(N+1));

disp('--- Hanning Window Coefficients ---');
disp(b_hann');

%% Frequency Response
figure(1);
freqz(b_hann,1);
title('Low-pass FIR using Hanning Window');

%% Stop Saving Command Window
diary off;

%% Save Figure (300 DPI)
exportgraphics(figure(1), 'Exp4_3_Hanning_Window.png', 'Resolution',300);

%% Completion Message
disp(' ');
disp('======================================');
disp('Experiment 4.3 Completed Successfully');
disp('Files Saved:');
disp('Exp4_3_Output.txt');
disp('Exp4_3_Hanning_Window.png');
disp('======================================');
```

Output

```
--- Hanning Window Coefficients ---
    0.0000
         0
   -0.0004
         0
    0.0013
         0
   -0.0028
         0
    0.0050
         0
   -0.0081
         0
    0.0122
         0
   -0.0179
         0
    0.0259
         0
```

```
-0.0378
      0
 0.0580
      0
-0.1027
      0
 0.3172
 0.5000
 0.3172
      0
-0.1027
      0
 0.0580
      0
-0.0378
      0
 0.0259
      0
-0.0179
      0
 0.0122
      0
-0.0081
      0
 0.0050
      0
-0.0028
      0
 0.0013
      0
-0.0004
      0
 0.0000
```

Output Image



Result: The FIR low-pass filter using the Hanning window was successfully designed and analyzed in MATLAB. The frequency response and phase response were obtained successfully.